Cybersecurity

Cybersecurity protects computer systems, networks, applications, and data from unauthorized access, attacks, and damage.

The CIA triad represents three important security principles: confidentiality, integrity, and availability.

Confidentiality ensures that information is accessible only to authorized users. Integrity protects information from unauthorized modification. Availability ensures that systems and data remain accessible when needed.

Encryption converts readable information into an encoded form using a cryptographic key. Authentication verifies the identity of a user or system.

Common cybersecurity threats include phishing, malware, ransomware, denial-of-service attacks, and password attacks.